

Table Of Content

Step 1: Data source for the Assignment 2.....	2
Step 2: SSAS Cube implementation.....	5
Step 3: Power BI Dashboard	5

Step 1: Data source for the Assignment 2

The data warehouse created in Assignment 1 was used as the data source for this assignment. It contains fact and dimension tables organized in a star schema, which supports analytical processing and reporting.

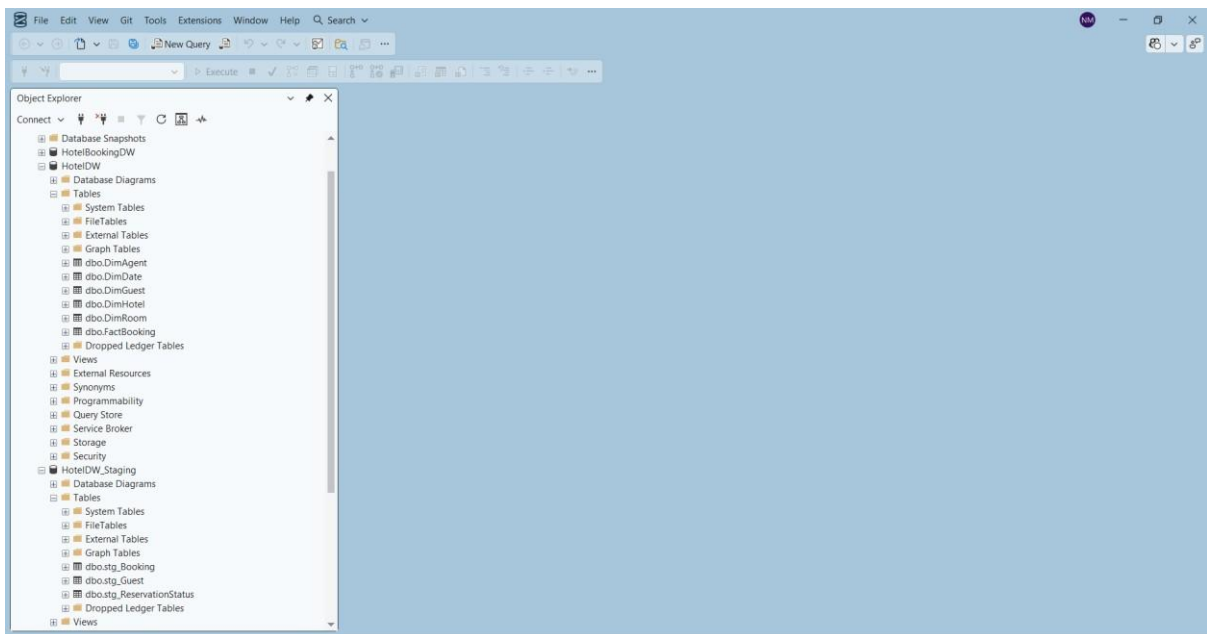
- Database Name : HotelDW
- Schema Type : Star Schema

Tables

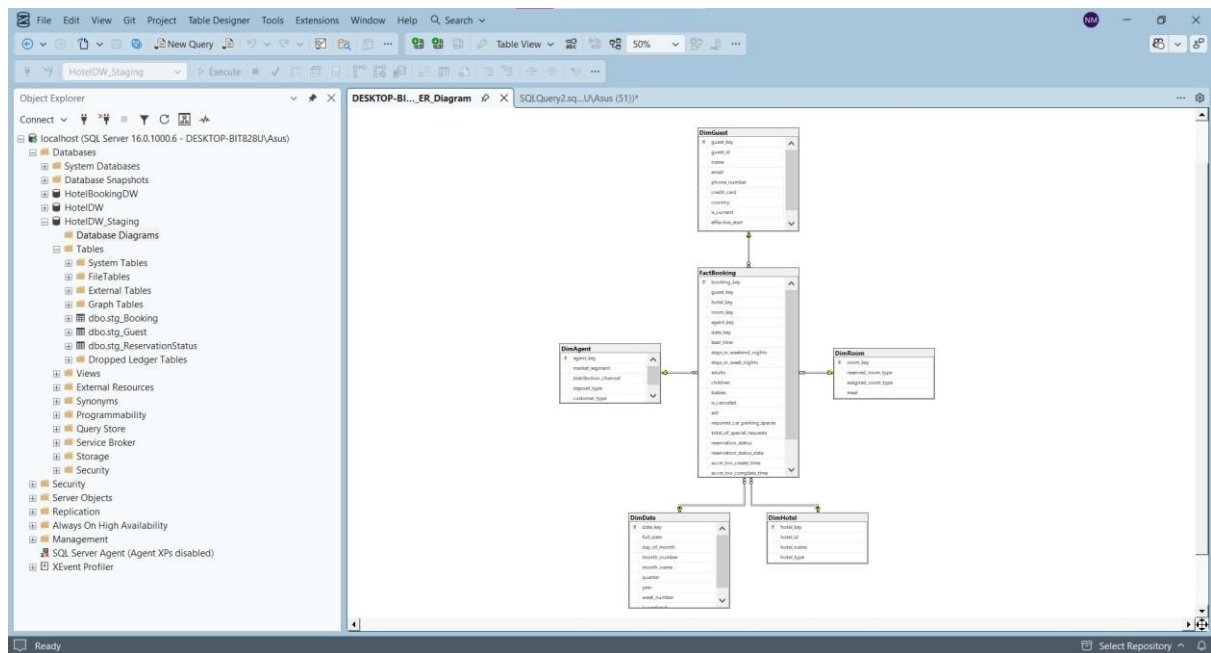
Table	Type	Description
FactBooking	Fact Table	Hotel booking transactions

DimGuest	Dimension	Guest/Customer details
DimRoom	Dimension	Room details
DimHotel	Dimension	Hotel details
DimAgent	Dimension	Booking agent details
DimDate	Dimension	Date details

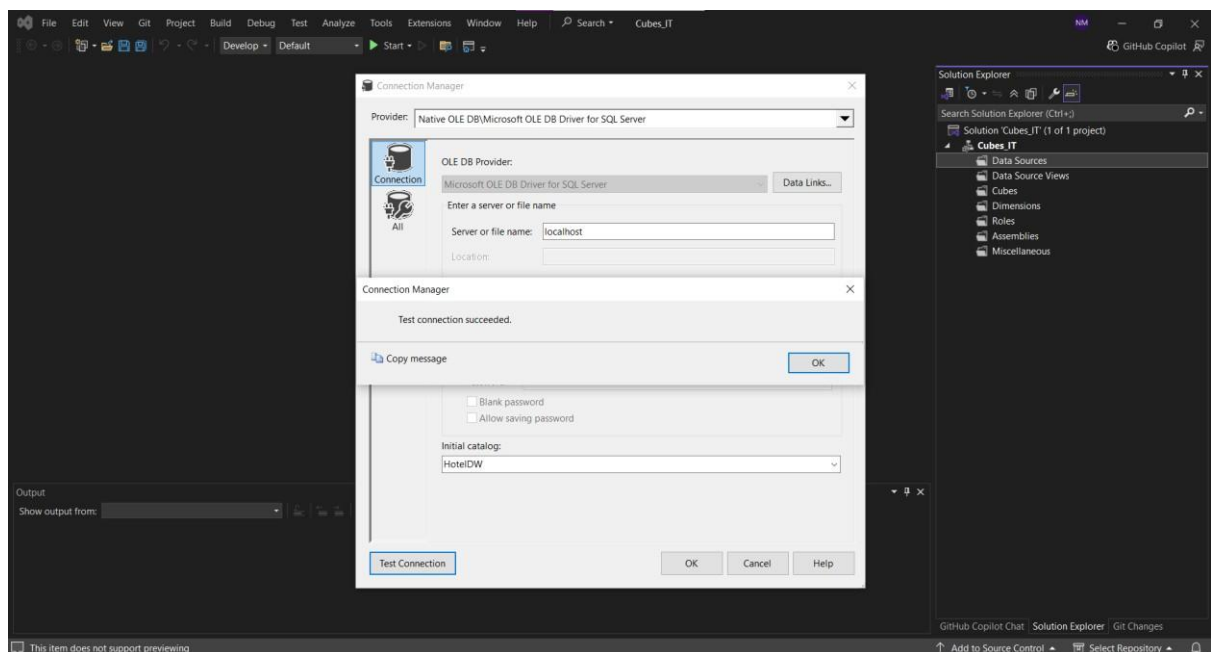
Database



ER Diagram



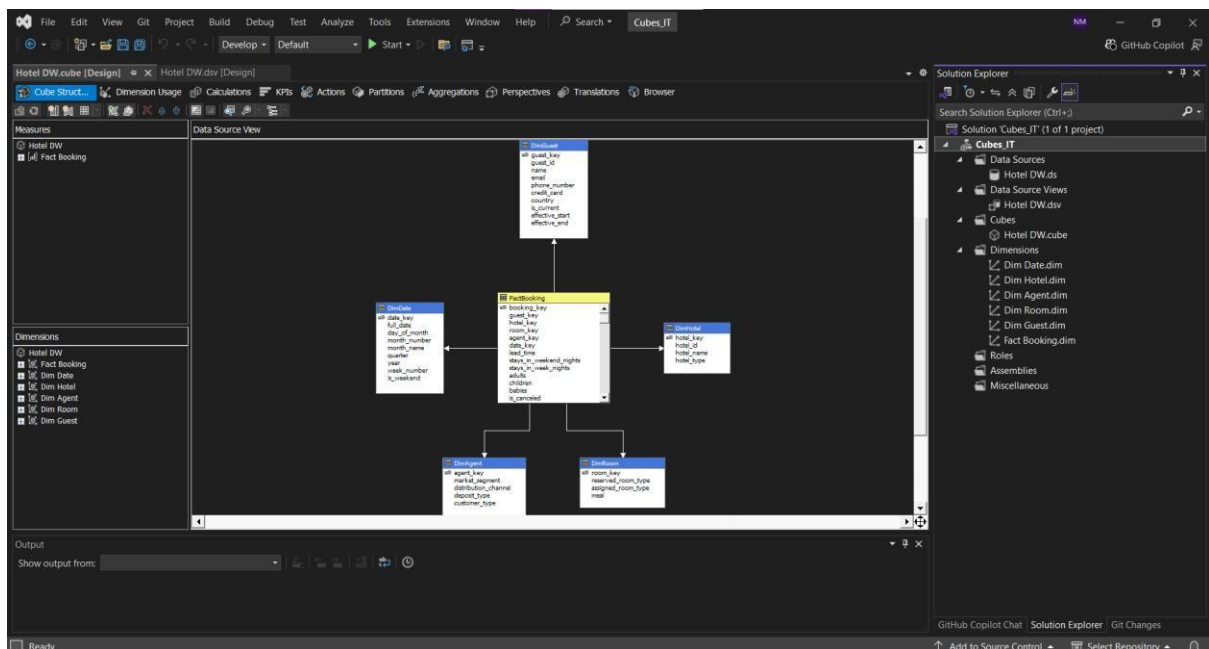
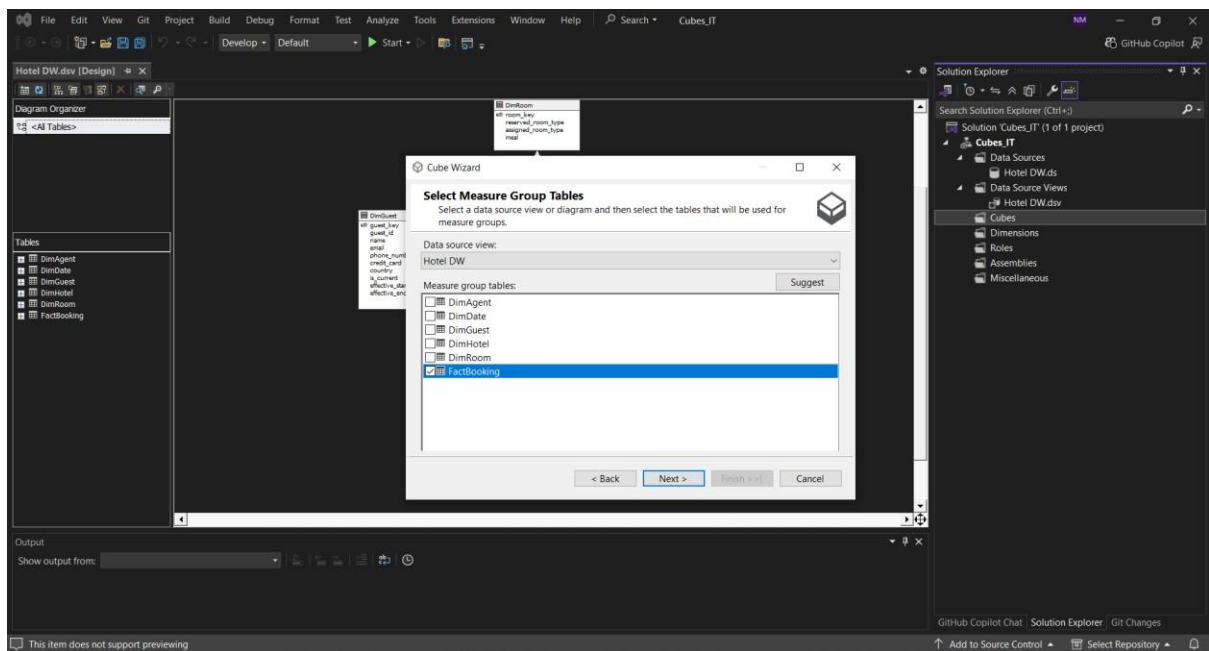
Data Source

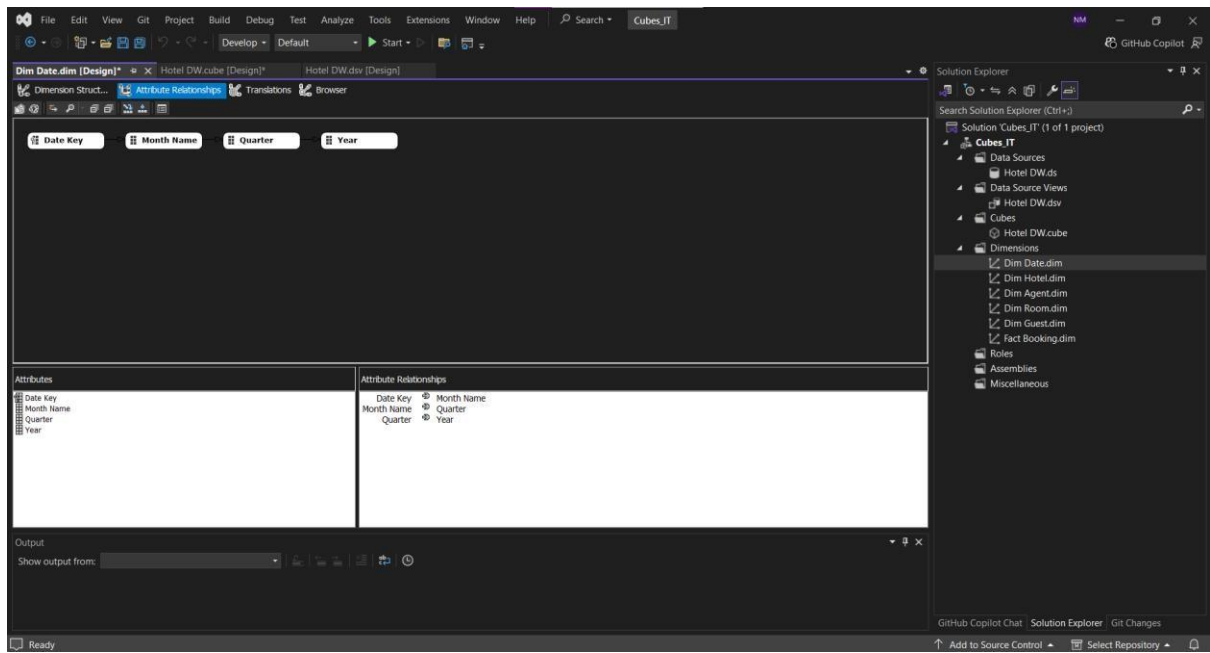
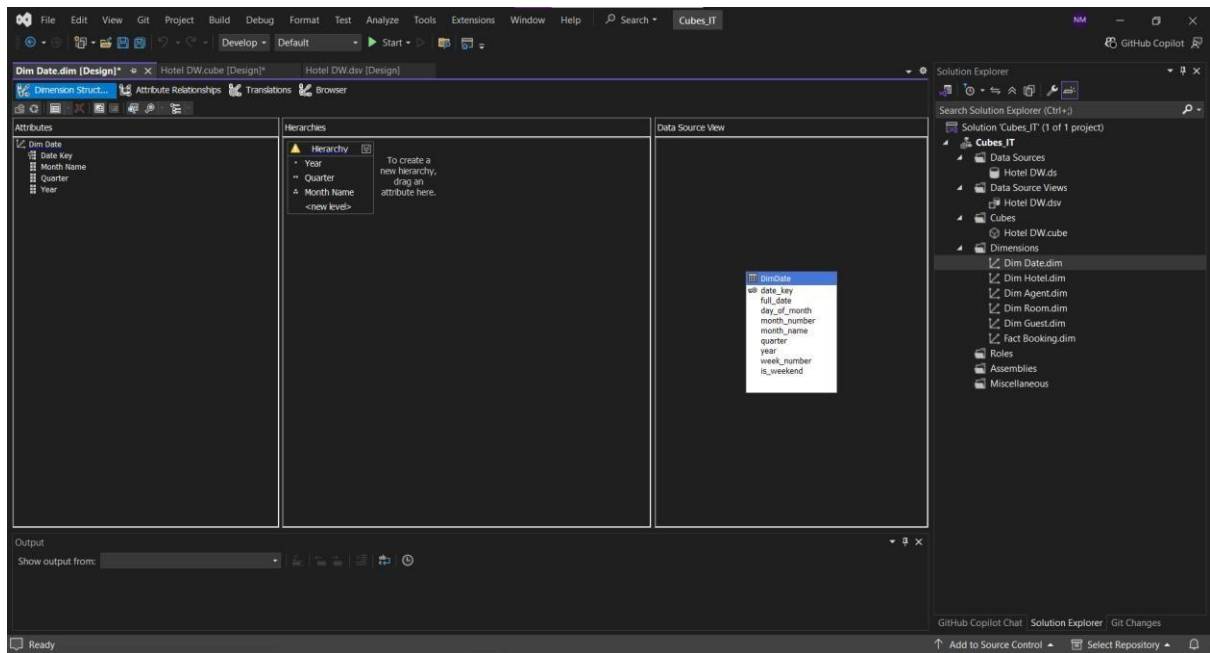


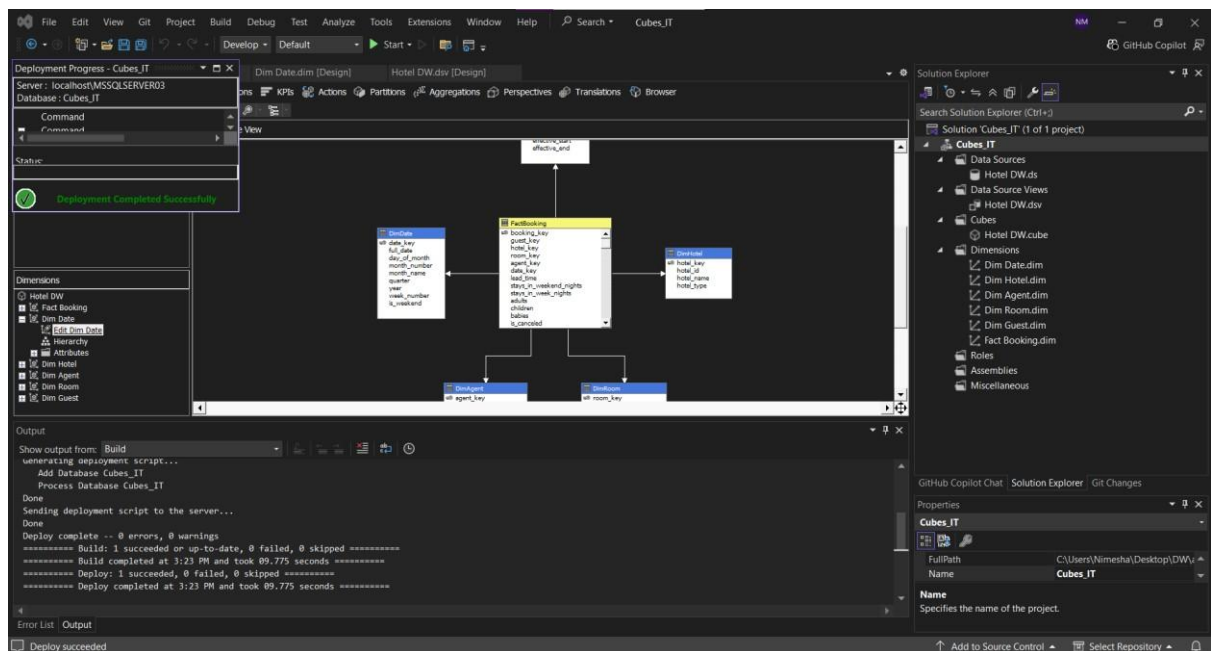
The screenshot displays the Microsoft SQL Server Enterprise Developer environment. The main window is the 'Data Source View Wizard', which is currently at the 'Select Tables and Views' step. The wizard prompts the user to 'Select objects from the relational database to be included in the data source view.' The 'Available objects' list on the left contains several tables from the 'dbo' schema, with 'DimAgent (dbo)' selected. The 'Included objects' list on the right is currently empty. The 'Filter' field is empty, and the 'Show system objects' checkbox is unchecked. The 'Add Related Tables' button is visible at the bottom right of the wizard. The 'Solution Explorer' on the right side of the screen shows the project structure, including 'Data Source Views' and 'Data Warehouse'. The 'Output' window at the bottom left is empty.



5







Step 3: Demonstration of OLAP operations & PowerBI Reports

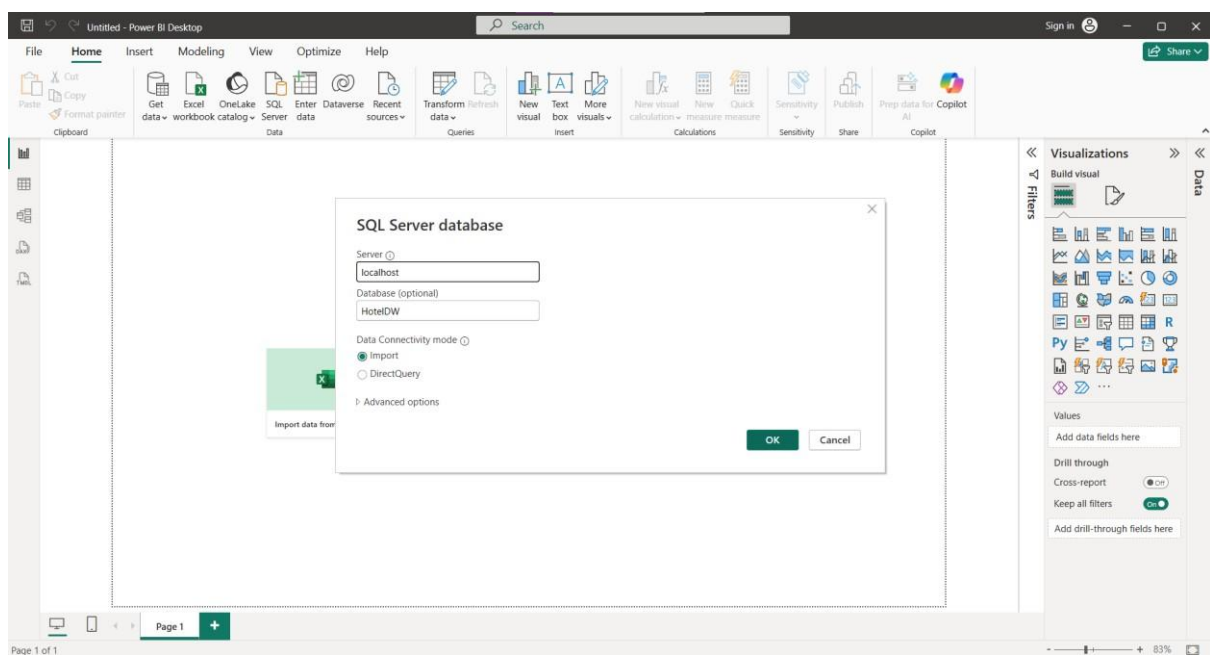
In this section, the Hotel DW OLAP cube deployed on SQL Server Analysis Services (SSAS) was connected to Microsoft Excel using the Data tab. A PivotTable was created by connecting to the local Analysis Services instance (localhost\MSSQLSERVER03) and selecting the Cubes_IT database with the Hotel DW cube. The following five OLAP operations were demonstrated using PivotTable configurations.

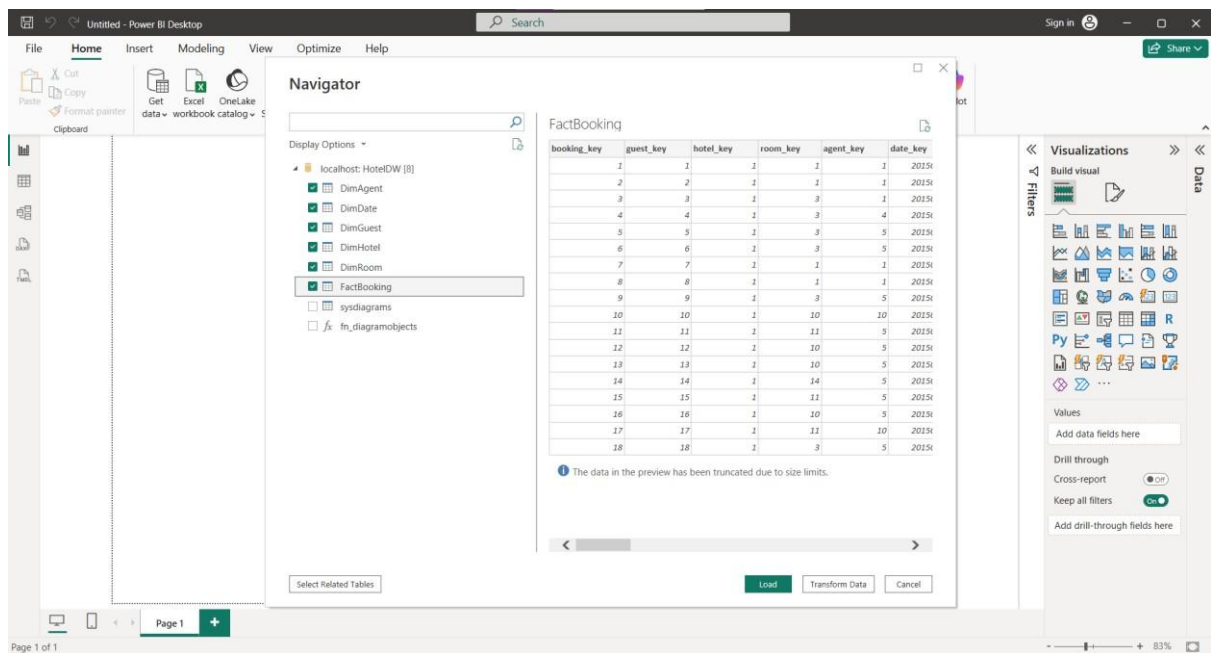
	A	B	C	D	E
1	Row Labels	Fact Booking Count			
2	2015	294262			
3	2016	754927			
4	2017	533721			
5	Grand Total	1582910			
6					
7					
8					
9					

PowerBI Reports

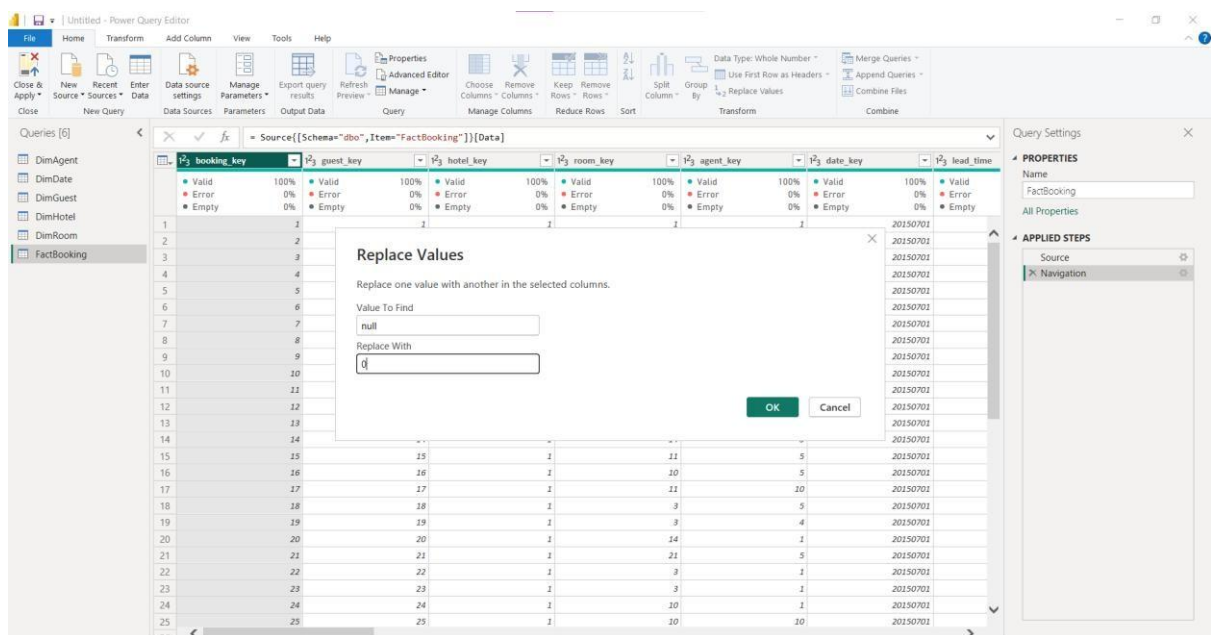
The data warehouse developed in Assignment 1 was utilized as the primary data source for building interactive business intelligence reports using Microsoft Power BI. The objective of this step is to transform structured warehouse data into meaningful visual insights that support decision-making in the hotel booking domain.

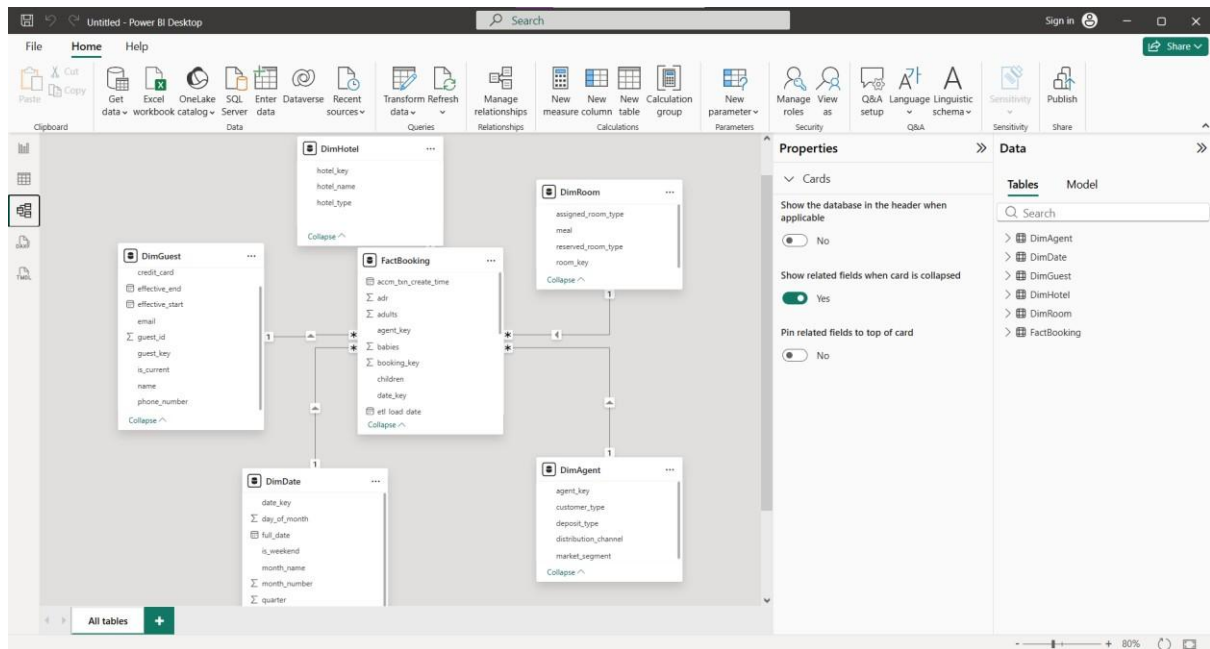
The Power BI reports are designed to demonstrate key analytical capabilities such as data summarization, filtering, hierarchical exploration, and detailed drill-down analysis.





Data Preparations



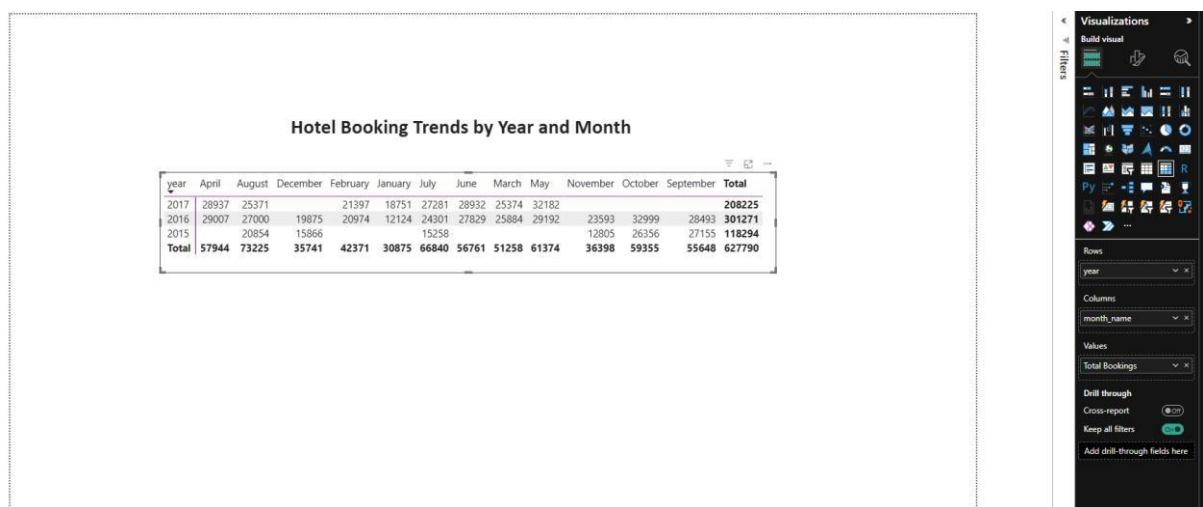


Report 1: Matrix Visual — Hotel Booking Summary

A new page named “**Matrix**” was created in Power BI Desktop. A Matrix visual was inserted from the Visualizations pane. The **Year** field from the DimDate table was placed in the **Rows** section to group data by year. The **Month_Name** field from the DimDate table was placed in the **Columns** section to provide a monthly breakdown of bookings. The **Total Bookings** measure, created using the COUNT of booking_key from the FactBooking table, was used in the **Values** section to display the total number of bookings for each year and month combination.

The Matrix visual displays booking distribution across months for each year, enabling detailed temporal analysis of booking trends. This structure allows comparison of booking patterns across different months within each year. The visual title was set using the Format Visual panel under General settings.

Hotel Booking Trends by Year and Month													
year	April	August	December	February	January	July	June	March	May	November	October	September	Total
2017	28937	25371		21397	18751	27281	28932	25374	32182				208225
2016	29007	27000	19875	20974	12124	24301	27829	25884	29192	23593	32999	28493	301271
2015		20854	15866			15258				12805	26356	27155	118294
Total	57944	73225	35741	42371	30875	66840	56761	51258	61374	36398	59355	55648	627790

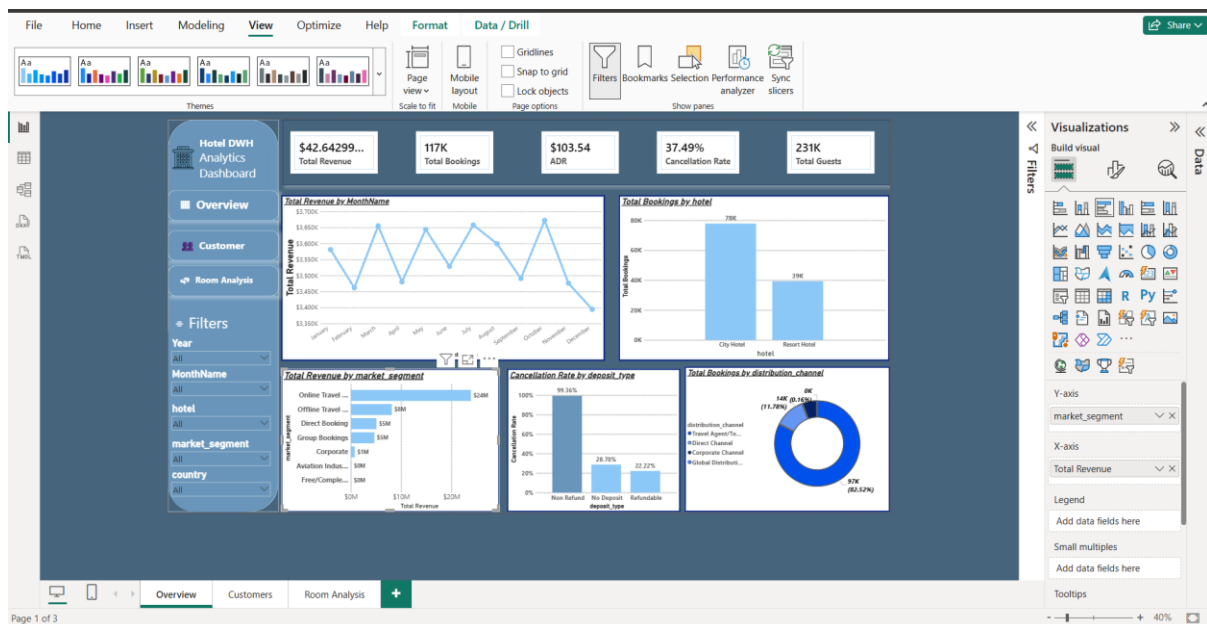


Report 2: Cascading Slicers with Multiple Visuals

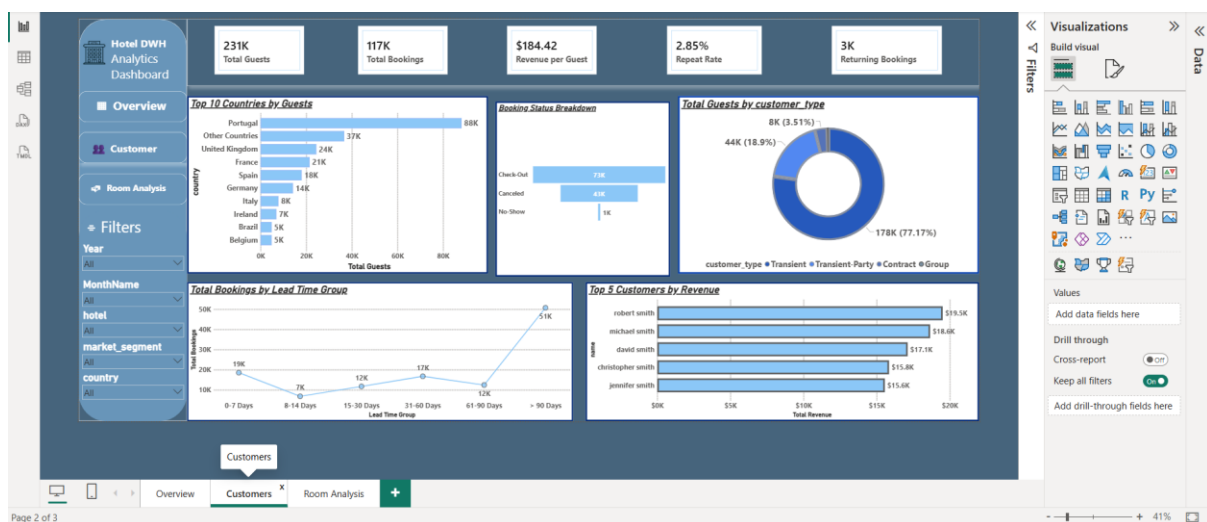
A new page named "Visuals" was created. The page title "Hotel Booking Analysis with Dynamic Filters" was added using a Text Box visual. Two Slicer visuals were added to the left side of the canvas. The first slicer was assigned the year field from DimDate with the title "Select Year", and the second slicer was assigned the quarter field from DimDate with the title "Select Quarter". To implement cascading filters, the year slicer was selected, Edit Interactions was enabled from the Format tab, and the filter interaction was set so that selecting a year value dynamically filters the available quarters in the second slicer.

A Clustered Bar Chart titled "Total Bookings by Hotel" was added to the right side of the canvas, with hotel_key on the Y-axis and Count of booking_key on the X-axis. A Line Chart titled "Booking Trend Over Years" was added below it, with year on the X-axis and Count of

booking_key on the Y-axis. Both visuals respond dynamically when values are selected in either slicer, enabling interactive filtering of the data.



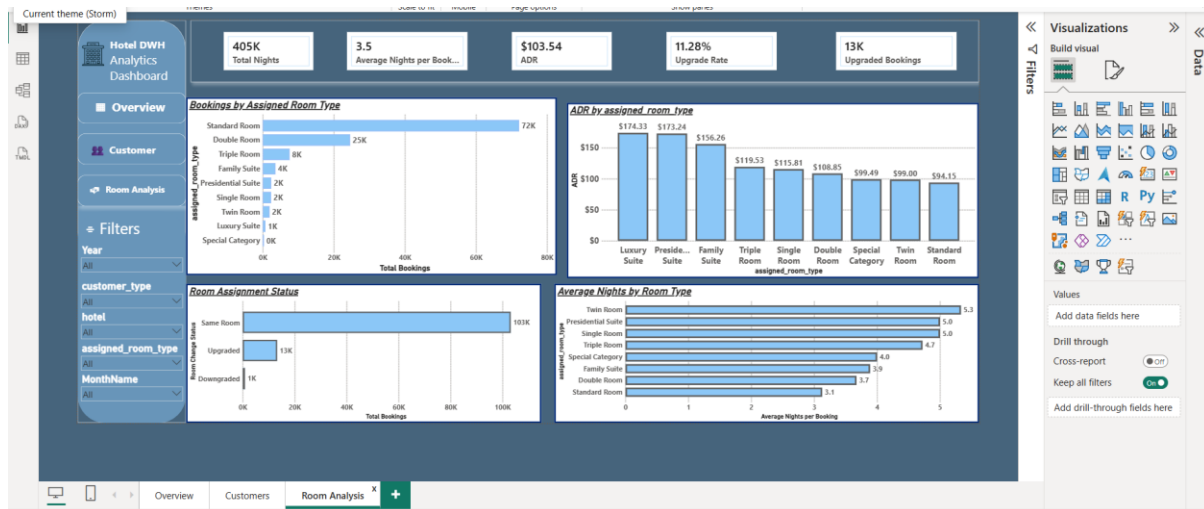
Report 3: Customers Overview



Publishing to Power BI Service

After completing the above four reports in Microsoft Power BI Desktop, the report file was saved and published to the Power BI Service using the publish option in the Home tab. After signing in with my Microsoft account, the appropriate workspace was selected and the report was successfully uploaded.

The published reports were then accessed through the Power BI Service (app.powerbi.com) to verify that all visuals, slicers, drill-down, and drill-through functionalities worked correctly in the online environment.



Conclusion

In this assignment, a complete BI solution was implemented starting from a data warehouse to SSAS cube, Excel OLAP analysis, and Power BI reporting. The system enables efficient data analysis and supports decision-making through interactive visualizations.